

## Subject Description Form

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| <b>Subject Code</b>                              | EIE4105                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Subject Title</b>                             | Multimodal Human Computer Interaction Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Credit Value</b>                              | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Level</b>                                     | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Pre-requisite</b>                             | EIE3103 Digital Signals and Systems;<br>EIE3124 Fundamentals of Machine Intelligence; or<br>EIE3312 Linear Systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Co-requisite/<br/>Exclusion</b>               | Nil                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Objectives</b>                                | This course aims at providing students with the theories and applications of multimodal human-computer interaction (HCI) technologies. In particular, it enables students to understand how machine learning and deep learning can be applied to various HCI systems.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Intended Subject Learning Outcomes</b>        | <p><b>Upon completion of the subject, students will be able to:</b></p> <p><u>Category A: Professional/academic knowledge and skills</u></p> <ol style="list-style-type: none"> <li>1. Understand the capability and benefits of various HCI technologies.</li> <li>2. Understand the theories of machine learning and deep learning.</li> <li>3. Understand how machine learning and deep learning can be applied to various HCI systems.</li> </ol> <p><u>Category B: Attributes for all-roundedness</u></p> <ol style="list-style-type: none"> <li>4. Understand the creative process when designing solutions to a problem.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Subject Synopsis/<br/>Indicative Syllabus</b> | <ol style="list-style-type: none"> <li>1. <u>HCI and Their Applications</u> <ol style="list-style-type: none"> <li>1.1 Applications of HCI in daily life</li> <li>1.2 Advantages of multimodal HCI</li> <li>1.3 Trends in HCI technologies</li> <li>1.4 Virtual reality, augmented reality, mixed reality, and metaverse</li> <li>1.5 Real-life examples of HCI</li> </ol> </li> <li>2. <u>Fundamental of Statistical Learning</u> <ol style="list-style-type: none"> <li>2.1 Probability and random variables</li> <li>2.2 Probability densities and distributions</li> <li>2.3 Sampling distributions</li> <li>2.4 Expectations and covariance</li> <li>2.5 Bayes rule and Bayes decision theory</li> <li>2.6 Curse of dimensionality</li> </ol> </li> <li>3. <u>Machine Learning for HCI</u> <ol style="list-style-type: none"> <li>3.1 Structure of pattern recognition systems.</li> <li>3.2 Unsupervised Learning: principal component analysis; Eigenface, K-means clustering; Gaussian mixture models; hidden Markov models</li> <li>3.3 Supervised Learning: linear discriminant analysis; support vector machines</li> </ol> </li> </ol> |

|                                                                                | <p>3.4 Deep Learning: deep neural networks; backpropagation; gradient-based optimization; convolutional neural networks; representation learning; deep learning development platforms</p> <p>3.5 Applications to handwriting recognition and face recognition.</p> <p>4. <u>Voice Computing</u></p> <p>4.1 Voice computing: Interacting with computer through voice</p> <p>4.2 Acoustic features</p> <p>4.3 HMM and DNN for speech recognition</p> <p>4.4 Language modelling</p> <p>4.5 Speaker recognition: GMM-UBM, GMM-SVM, i-vectors, x-vectors, DNN speaker embedding, LDA, and PLDA</p> <p>4.6 Applications of voice computing: voice search, spoken dialog systems, natural language processing, speech emotion recognition, speaker recognition, voice cloning.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                   |             |                                                                                |   |  |  |   |   |   |   |                                       |  |  |  |  |  |                            |     |   |   |   |   |                     |     |   |   |   |  |                        |     |  |  |   |   |                |     |   |   |   |   |       |      |  |  |  |  |
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| <b>Teaching/Learning Methodology</b>                                           | <p>Lectures: The subject matters will be delivered through lectures. Students will be engaged in the lectures through Q&amp;A, discussions, and specially designed classroom activities.</p> <p>Tutorials: During tutorials, students will work on/discuss some chosen topics. This will help strengthen the knowledge taught in lectures.</p> <p>Laboratory and assignments: During laboratory exercises, students will perform hands-on tasks to practice what they have learned. They will evaluate performance of systems and design solutions to problems. The assignments will help students to review the knowledge taught in class.</p> <p>While lectures and tutorials will help to achieve the professional outcomes, the open-ended questions in laboratory exercises and assignments will provide the chance to students to exercise their creativity in problem solving.</p>                                                                                                                                                                                                                                                                                                                                                                                           |                                   |             |                                                                                |   |  |  |   |   |   |   |                                       |  |  |  |  |  |                            |     |   |   |   |   |                     |     |   |   |   |  |                        |     |  |  |   |   |                |     |   |   |   |   |       |      |  |  |  |  |
| <b>Assessment Methods in Alignment with Intended Subject Learning Outcomes</b> | <table><tr><th rowspan="2">Specific Assessment Methods/Tasks</th><th rowspan="2">% Weighting</th><th colspan="4">Intended Subject Learning Outcomes to be Assessed (Please tick as appropriate)</th></tr><tr><th>1</th><th>2</th><th>3</th><th>4</th></tr><tr><td>1. Continuous Assessment (total: 50%)</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>• Homework and assignments</td><td>15%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>• Tests and Quizzes</td><td>20%</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr><tr><td>• Laboratory exercises</td><td>15%</td><td></td><td></td><td>✓</td><td>✓</td></tr><tr><td>2. Examination</td><td>50%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Total</td><td>100%</td><td colspan="4"></td></tr></table> <p><b>Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:</b></p> <p>Assignment, homework, and laboratory exercises will require students to apply what they have learnt to solve problems. There will be open-ended questions that allow students to exercise their creativity in making design.</p> <p>Examination and tests: They assess students’ achievement of the learning outcomes in a more formal manner.</p> | Specific Assessment Methods/Tasks | % Weighting | Intended Subject Learning Outcomes to be Assessed (Please tick as appropriate) |   |  |  | 1 | 2 | 3 | 4 | 1. Continuous Assessment (total: 50%) |  |  |  |  |  | • Homework and assignments | 15% | ✓ | ✓ | ✓ | ✓ | • Tests and Quizzes | 20% | ✓ | ✓ | ✓ |  | • Laboratory exercises | 15% |  |  | ✓ | ✓ | 2. Examination | 50% | ✓ | ✓ | ✓ | ✓ | Total | 100% |  |  |  |  |
| Specific Assessment Methods/Tasks                                              | % Weighting                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                   |             | Intended Subject Learning Outcomes to be Assessed (Please tick as appropriate) |   |  |  |   |   |   |   |                                       |  |  |  |  |  |                            |     |   |   |   |   |                     |     |   |   |   |  |                        |     |  |  |   |   |                |     |   |   |   |   |       |      |  |  |  |  |
|                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 1                                 | 2           | 3                                                                              | 4 |  |  |   |   |   |   |                                       |  |  |  |  |  |                            |     |   |   |   |   |                     |     |   |   |   |  |                        |     |  |  |   |   |                |     |   |   |   |   |       |      |  |  |  |  |
| 1. Continuous Assessment (total: 50%)                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                   |             |                                                                                |   |  |  |   |   |   |   |                                       |  |  |  |  |  |                            |     |   |   |   |   |                     |     |   |   |   |  |                        |     |  |  |   |   |                |     |   |   |   |   |       |      |  |  |  |  |
| • Homework and assignments                                                     | 15%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | ✓                                 | ✓           | ✓                                                                              | ✓ |  |  |   |   |   |   |                                       |  |  |  |  |  |                            |     |   |   |   |   |                     |     |   |   |   |  |                        |     |  |  |   |   |                |     |   |   |   |   |       |      |  |  |  |  |
| • Tests and Quizzes                                                            | 20%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | ✓                                 | ✓           | ✓                                                                              |   |  |  |   |   |   |   |                                       |  |  |  |  |  |                            |     |   |   |   |   |                     |     |   |   |   |  |                        |     |  |  |   |   |                |     |   |   |   |   |       |      |  |  |  |  |
| • Laboratory exercises                                                         | 15%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                   |             | ✓                                                                              | ✓ |  |  |   |   |   |   |                                       |  |  |  |  |  |                            |     |   |   |   |   |                     |     |   |   |   |  |                        |     |  |  |   |   |                |     |   |   |   |   |       |      |  |  |  |  |
| 2. Examination                                                                 | 50%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | ✓                                 | ✓           | ✓                                                                              | ✓ |  |  |   |   |   |   |                                       |  |  |  |  |  |                            |     |   |   |   |   |                     |     |   |   |   |  |                        |     |  |  |   |   |                |     |   |   |   |   |       |      |  |  |  |  |
| Total                                                                          | 100%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                   |             |                                                                                |   |  |  |   |   |   |   |                                       |  |  |  |  |  |                            |     |   |   |   |   |                     |     |   |   |   |  |                        |     |  |  |   |   |                |     |   |   |   |   |       |      |  |  |  |  |

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| <b>Student Study Effort Expected</b> | <b>Class contact (time-tabled):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                  |
|                                      | • Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 24 Hours         |
|                                      | • Tutorial/Laboratory/Practice Classes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 15 Hours         |
|                                      | <b>Other student study effort:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                  |
|                                      | • Lecture: preview/review of notes; homework/assignment; preparation for test/quizzes/examination                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 36 Hours         |
|                                      | • Tutorial/Laboratory/Practice Classes: preview of materials, revision and/or reports writing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 30 Hours         |
|                                      | <b>Total student study effort:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>105 Hours</b> |
| <b>Reading List and References</b>   | <b>Reference Materials:</b> <ol style="list-style-type: none"> <li>1. M.W. Mak and J.T. Chien, <i>Machine Learning for Speaker Recognition</i>, Cambridge University Press, 2020.</li> <li>2. I. Goodfellow, Y. Bengio and A. Courville, <i>Deep Learning</i>, MIT Press, 2016.</li> <li>3. S.Y. Kung, M.W. Mak and S.H. Lin, <i>Biometric Authentication: A Machine Learning Approach</i>, Prentice Hall, 2005.</li> <li>4. R. Haeb-Umbach, <i>et al.</i> "Speech Processing for Digital Home Assistants: Combining Signal Processing with Deep-learning Techniques", <i>IEEE Signal Processing Magazine</i>, Nov. 2019.</li> <li>5. C.M. Bishop, <i>Pattern Recognition and Machine Learning</i>, Springer, 2006.</li> <li>6. S.J.D. Prince, <i>Computer Vision: Models, Learning, and Inference</i>, Cambridge University Press, 2012.</li> <li>7. J.P. Thiran, F. Marques and H. Bourlard, <i>Multimodal Signal Processing, Theory and Applications for Human Computer Interaction</i>, Elsevier, 2010.</li> <li>8. S. Greengard, <i>Virtual Reality</i>, MIT Press Essential Knowledge Series, 2019.</li> </ol> |                  |